

Module-2

2.

8 M
5 W

 M M M M M M M M

$8M \rightarrow \text{ways} = 8!$

gaps = 9 $\therefore$ 5 w ways = 9C_5
and internal arrangement
= $5!$

$$\therefore \text{Ans.} = {}^9C_5 \times 5! \times 8! \\ = 609638400$$

3.

4 IT
17 CSE

$$17 \text{ CS}_6 \rightarrow 17 \text{f}$$

$$\therefore 4 \text{ IT} \rightarrow \text{ways} = {}^{18-4+1}C_4 \times 4!$$

Choose r positions
from n positions such
that no 2 chosen
positions are
consecutive —

ways = $n-r+1 C_r$

4. Rank of the word 'GOOGLE'



$$E \text{ -----} \rightarrow \frac{5!}{2!2!} = \frac{5 \times 4 \times 3}{2} = 30$$

$$GE \text{ -----} \rightarrow \frac{4!}{2!} = 4 \times 3 = 12$$

$$GG \text{ -----} \rightarrow \frac{4!}{2!} = 4 \times 3 = 12$$

$$GL \text{ -----} \rightarrow \frac{4!}{2!} = 4 \times 3 = 12$$

$$GOO \text{ ----} \rightarrow 3! = 3 \times 2 = 6$$

$$GOOE \text{ --} \rightarrow 2! = 2$$

$$GOOGE \text{ -} \rightarrow 1! = 1$$

$$GOOGLE \text{ } \rightarrow 1$$

$$30 + 12 + 12 + 12 + 6 + 2 + 1 + 1 = 78$$

5. 'MODESTY' - Rank?

$$D \text{ -----} \rightarrow 6! = 720$$

$$E \text{ -----} \rightarrow 6! = 720$$

$$MD \text{ -----} \rightarrow 5! = 120$$

$$ME \text{ -----} \rightarrow 5! = 120$$

$$MODESTY \rightarrow 1$$

$$720 + 720 + 120 + 120 + 1 = 1681$$

6. $5x + 4y + 5z + 7w$ let, $x+y+z+w=10$

$$0 \leq x \leq 5, 0 \leq y \leq 4, 0 \leq z \leq 5, 0 \leq w \leq 7$$

H non -ve integer solns are.

$$= {}^{n+r-1}C_{r-1} \quad n=10, r=4$$

$$= {}^{10+4-1}C_{4-1} = {}^{13}C_3 = 286$$

Invalid solns

case 1: $x \geq 6$

$$\text{let, } x' = x - 6 \therefore x' + y + z + w = 10 - 6 = 4$$

$$\# \text{ soln} = 4+4-1C_{4-1} = 7C_3 = 35$$

1. Non-negative solutions (≥ 0)

$$x_1 + x_2 + \dots + x_r = n$$

$$\binom{n+r-1}{r-1}$$

2. Positive solutions (≥ 1)

$$x_1 + x_2 + \dots + x_r = n$$

$$\binom{n-1}{r-1}$$

because each variable gets at least 1 first.

